

JOKIOINEN, Finland

WMO: 029630

Lat: 60.814N

Lon: 23.497E

Elev: 338

StdP: 14.52

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.7	-3.5	-13.9	2.6	-9.5	-7.6	3.7	-3.0	21.6	28.3	19.7	28.2	4.2	60	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	80.1	64.6	76.9	63.0	73.6	61.6	67.4	75.7	65.4	73.4	63.4	70.9	8.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.5	91.9	70.9	62.3	85.0	68.6	60.3	79.0	66.4	32.0	75.8	30.4	73.7	28.9	71.1	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	17.5	15.7	-15.0	83.8	6.4	4.0	-19.6	86.7	-23.4	89.0	-27.0	91.3	-31.7	94.2		
(5)			-15.3	69.3	6.4	2.5	-19.9	71.2	-23.6	72.6	-27.2	74.1	-31.8	75.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	22.0	21.4	27.9	39.1	49.2	57.0	62.4	59.8	51.4	41.0	33.0	26.8		
	DBStd	16.35	11.36	11.76	8.35	6.76	7.29	5.87	5.36	5.42	6.07	7.09	8.11	10.39		
	HDD50	4371	867	800	686	333	104	7	0	2	56	287	509	720		
	HDD65	8811	1332	1220	1151	777	493	247	118	176	409	744	959	1185		
	CDD50	1097	0	0	0	7	78	219	383	306	97	7	0	0		
	CDD65	62	0	0	0	0	2	9	36	15	0	0	0	0		
	CDH74	631	0	0	0	0	34	108	362	127	0	0	0	0		
	CDH80	92	0	0	0	0	3	10	66	13	0	0	0	0		
	WSAvg	7.8	8.1	8.0	8.3	7.9	7.9	7.7	7.0	7.1	7.3	8.0	8.1	8.5		
	PrecAvg	23.3	1.6	1.1	1.1	1.2	1.4	2.3	3.0	3.0	2.3	2.4	2.2	1.8		
(15) Precipitation	PrecMax	31.0	3.2	2.9	2.0	3.3	3.4	5.2	6.5	7.2	5.4	6.1	5.0	3.6		
	PrecMin	14.7	0.2	0.0	0.0	0.1	0.0	0.4	0.0	0.5	0.5	0.5	0.1	0.0		
	PrecStd	3.4	0.7	0.6	0.5	0.7	0.7	1.2	1.5	1.4	1.2	1.2	1.0	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	41.2	50.9	66.8	78.2	81.3	85.6	81.7	70.6	57.7	49.3	45.7		
		MCWB	39.5	38.0	42.0	51.7	59.7	63.9	67.0	65.4	60.1	52.9	48.2	44.5		
	2%	DB	38.1	37.8	44.7	61.2	73.1	77.2	81.7	77.3	67.2	54.8	47.3	43.1		
		MCWB	37.1	36.3	38.4	48.6	57.3	61.9	66.3	63.4	58.6	52.2	46.1	42.2		
	5%	DB	36.3	36.0	40.9	56.2	68.6	73.4	78.2	74.3	63.8	53.0	45.3	40.7		
		MCWB	35.6	34.9	36.4	46.1	55.1	60.3	64.7	62.4	56.5	51.1	44.4	39.7		
	10%	DB	34.5	34.6	37.7	51.9	63.8	69.7	74.5	70.8	61.0	51.2	43.5	38.4		
		MCWB	33.9	33.6	34.7	43.7	52.2	58.0	63.2	60.9	55.9	49.5	42.6	37.6		
	0.4%	WB	39.9	38.4	42.9	53.4	63.5	67.4	71.0	68.4	62.2	54.8	48.6	44.8		
		MCDB	40.2	39.9	49.3	63.1	73.8	76.0	79.9	76.6	66.9	56.1	49.2	45.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	36.7	39.3	49.9	59.6	64.4	68.5	66.1	60.1	53.1	46.3	42.4		
		MCDB	37.8	37.5	43.3	58.9	68.8	74.0	77.1	73.3	64.6	54.3	46.9	43.0		
	5%	WB	35.7	35.2	37.1	47.1	56.5	61.7	66.4	64.1	58.4	51.4	44.5	39.9		
		MCDB	36.2	35.9	39.8	54.7	65.5	70.0	74.9	70.9	62.1	52.7	45.3	40.5		
	10%	WB	34.0	33.7	35.3	44.5	53.4	59.4	64.5	62.2	56.7	49.7	42.7	37.6		
		MCDB	34.4	34.4	37.3	51.1	62.5	67.5	72.5	68.5	60.1	51.0	43.3	38.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.4	13.5	16.0	18.9	16.9	17.2	17.2	14.3	10.0	7.0	8.2		
		MCDBR	6.7	8.8	15.9	24.4	26.4	23.6	23.7	22.2	18.2	10.6	7.3	7.0		
		MCWBR	6.6	8.0	10.8	13.8	13.0	11.2	10.5	10.1	10.5	8.0	7.1	6.7		
		MCDBR	6.9	8.0	13.6	21.8	22.0	19.7	20.0	17.7	15.0	9.1	7.1	6.6		
	5% WB	MCWBR	6.9	7.5	9.8	13.0	11.9	10.8	9.9	9.4	9.7	7.7	7.1	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.300	0.326	0.363	0.359	0.354	0.379	0.369	0.341	0.325	0.288	0.377		
	taud		2.108	2.182	2.265	2.344	2.416	2.451	2.398	2.422	2.493	2.407	2.202	2.634		
	Ebn at Noon		149	213	251	264	275	278	266	260	248	208	138	102		
	Edh at Noon		13	22	29	32	32	32	33	30	23	18	12	9		
(44) All-Sky Solar Radiation	RadAvg		74	255	727	1205	1640	1726	1624	1276	778	337	92	43		
	RadStd		13	54	89	127	165	139	163	137	99	59	17	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+0.86	N/A	N/A	N/A	N/A	N/A	-265	N/A	N/A	N/A	
(47)	Regional (59 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	-243	N/A	N/A	N/A	(47)

Nomenclature: See separate page